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Paul A. Yushkevich, Ph.D.

Professor of Radiology

Department: Radiology

Graduate Group Affiliations

- Bioengineering

Contact Information

Penn Image Computing and Science Laboratory

Richards Building, 6th Floor, D605

3700 Hamilton Walk

Philadelphia, PA 19104

Office: 215-746-6924

Email: pauly2@pennmedicine.upenn.edu

Publications

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Education

B.S. (Computer Science And Mathematics) University Of North Carolina At Charlotte, 1996. M.S. (Computer Science) University Of North Carolina Chapel Hill, 2000. Ph.D. (Computer Science) University Of North Carolina Chapel Hill, 2003.

Description of Research Expertise

My research portfolio combines theoretical work on statistical shape characterization using a symmetry-based representation of shape with applied and translational research in multiple areas of biomedical image analysis. I am particularly interested in analysis techniques that are tailored to specific anatomical structures. My key work in this area involves automatic segmentation and morphometry of the hippocampal formation (HF) in magnetic resonance imaging (MRI). The HF plays a central role in memory function and is a site of early neurodegeneration in Alzheimer's disease. In a broad effort to develop more detailed imaging-based biomarkers for Alzheimer's disease, my team has developed a first of its kind detailed computational atlas of the HF from postmortem MRI microscopy. With NIH R01 funding, we are now integrating this atlas with histology, leveraging it for the analysis of the HF and its subfields in in vivo MRI, and evaluating MRI-derived subfield-specific as alternative imaging biomarkers in dementia. We recently published a pioneering technique for automatic segmentation of HF subfields in high-resolution, HF-focused T2-weighted in vivo MRI, showing excellent agreement with manual segmentation. Concurrently, we developed and evaluated a strategy for improving the accuracy of general-purpose segmentation techniques using machine learning as a corrective wrapper method. Other recent publications include a surface-based tract-specific strategy for diffusion MRI analysis, for which open-source software is currently being developed, and an automatic cardiac MRI segmentation method with an explicit prior on heart wall thickness.

Selected Publications

Buehner BJ, Morcillo-Nieto AO, Zsadanyi SE, Rozalem Aranha M, Arriola-Infante JE, Vaqué-Alcázar L, Arranz J, Rodríguez-Baz Í, Maure Blesa L, Videla L, Barroeta I, Del Hoyo Soriano L, Benejam B, Fernández S, Sanjuan Hernandez A, Bargalló N, González-Ortiz S, Giménez S, de Flores R, Yushkevich PA, Alcolea D, Belbin O, Lleó A, Carmona-Iragui M, Fortea J, Bejanin A.: *Medial temporal lobe atrophy in Down syndrome along the Alzheimer's disease continuum*. Brain 148: 2509-2521, Jul 2025.

Athalye C, Bahena A, Khandelwal P, Emrani S, Trotman W, Levorse LM, Khodakarami Z, Ohm DT, Teunissen-Bermeo E, Capp N, Sadaghiani S, Arezoumandan S, Lim SA, Prabhakaran K, Ittyerah R, Robinson JL, Schuck T, Lee EB, Tisdall MD, Das SR, Wolk DA, Irwin DJ, Yushkevich PA.: *Operationalizing postmortem pathology-MRI association studies in Alzheimer's disease and related disorders with MRI-guided histology sampling*. Acta Neuropathol Commun 13: 120, May 2025.

Duong MT, Das SR, Khandelwal P, Lyu X, Xie L, McGrew E, Dehghani N, McMillan CT, Lee EB, Shaw LM, Yushkevich PA, Wolk DA, Nasrallah IM.: *Hypometabolic mismatch with atrophy and tau pathology in mixed Alzheimer's and Lewy body disease*. Brain 148: 1577-1587, May 2025.

Cohen JS, Phillips J, Das SR, Olm CA, Radhakrishnan H, Rhodes E, Cousins KAQ, Xie SX, Nasrallah IM, Yushkevich PA, Wolk DA, Lee EB, Weintraub D, Irwin DJ, McMillan CT.: *Posterior hippocampal sparing in Lewy body disorders with Alzheimer's copathology: An in vivo MRI study*. Neuroimage Clin 2025.

Xie L, Das SR, Li Y, Wisse LEM, McGrew E, Lyu X, DiCalogero M, Shah U, Ilesanmi A, Denning AE, Brown CA, Cohen J, Sreepada L, Dong M, Sadeghpour N, Khandelwal P, Ittyerah R, Ravikumar S, Sadaghiani S, Hrybouski S, de Flores R, Gibson E, Yushkevich PA, Wolk DA; Alzheimer's Disease Neuroimaging Initiative.: *A multi-cohort study of longitudinal and cross-sectional Alzheimer's disease biomarkers in cognitively unimpaired older adults*. Alzheimers Dement 2025.

Brown CA, Das SR, Cousins KAQ, Tropea TF, Plotkin AC, Detre JA, Yushkevich PA, McMillan CT, Lee EB, Shaw LM, Nasrallah IM; Alzheimer's Disease Neuroimaging Initiative; Wolk DA.: *Tau burden is best captured by magnitude and extent: Tau-MaX as a measure of global tau*. Alzheimers Dement 2025.

Wuestefeld A, Xie L, McGrew E, Pichet-Binette A, Spotorno N, van Westen D, Mattsson-Carlgren N, Yushkevich PA, Das SR, Wolk DA, Wisse LEM; and for the Alzheimer's Disease Neuroimaging Initiative.: *Tau, atrophy, and domain-specific cognitive impairment in typical Alzheimer's disease*. Alzheimers Dement 2025.

Hrybouski S, Das SR, Xie L, Brown CA, Flamporis M, Lane J, Nasrallah IM, Detre JA, Yushkevich PA, Wolk DA.: *BOLD amplitude correlates of preclinical Alzheimer's disease*. Neurobiol Aging 2025.

Sreepada LP, Brown CA, Das SR, Yushkevich PA, Wolk DA, McMillan CT; Alzheimer's Disease Neuroimaging Initiative.: *Biological age acceleration in Alzheimer's disease modulates relative cortical to medial temporal lobe neurodegeneration*. Neurobiol Aging 2025.

Sadeghpour N, Lim SA, Wuestefeld A, Denning AE, Ittyerah R, Trotman W, Chung E, Sadaghiani S, Prabhakaran K, Bedard ML, Ohm DT, Artacho-Pérula E, de Onzoño Martin MMI, Muñoz M, Romero FJM, González JCD, Del Arroyo Jiménez M, Del Marcos Rabal M, Insausti Serrano AM, González NV, Sánchez SC, de la Rosa Prieto C, Insausti R, McMillan C, Lee EB, Detre JA, Das SR, Xie L, Tisdall MD, Irwin DJ, Wolk DA, Yushkevich PA, Wisse LEM.: *Developing an Anatomically Valid Segmentation Protocol for Anterior Regions of the Medial Temporal Lobe for Neurodegenerative Diseases*. Hippocampus 2025.

Contact and Directions

Perelman School of Medicine
University of Pennsylvania
3400 Civic Center Boulevard
Building 421, Philadelphia, PA 19104

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